



23/3/16

①

Unit-3

Sequential MOS logic circuit: SR latch, clocked latch and flip-flop circuits, CMOS D latch and edge trigger flip-flop, dynamic logic circuits; basic principle, non-ideal effects, domino CMOS logic, high performance dynamic CMOS ckt, clocking issues, clock distribution.

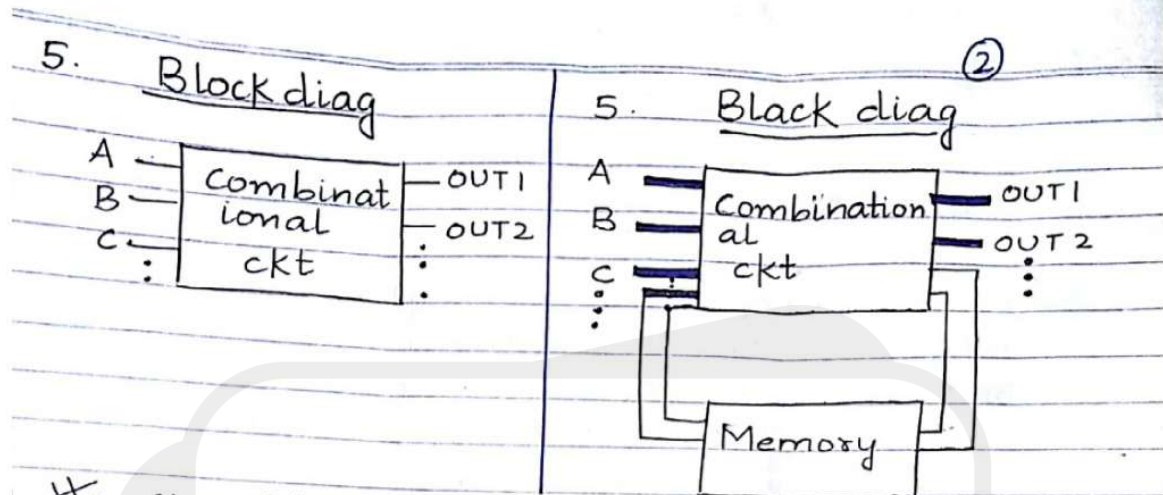
Difference b/w combinational & sequential ckt

<u>Combinational ckt</u>	<u>Sequential ckt</u>
1. O/p depends on <u>present input</u>	1. O/p depends on <u>present input</u> and <u>previously applied input variables</u>
2. Feedback is <u>absent</u>	2. Feedback is <u>present</u> .
3. <u>No storage element</u> is present.	3. <u>Storage element</u> is present.
4. They are classified as <u>non-regenerative ckt</u> since there is no feedback relationship b/w i/p and o/p.	4. They are classified as <u>regenerative ckt</u> since there is a direct or indirect ^{feedback} relationship b/w i/p and o/p.

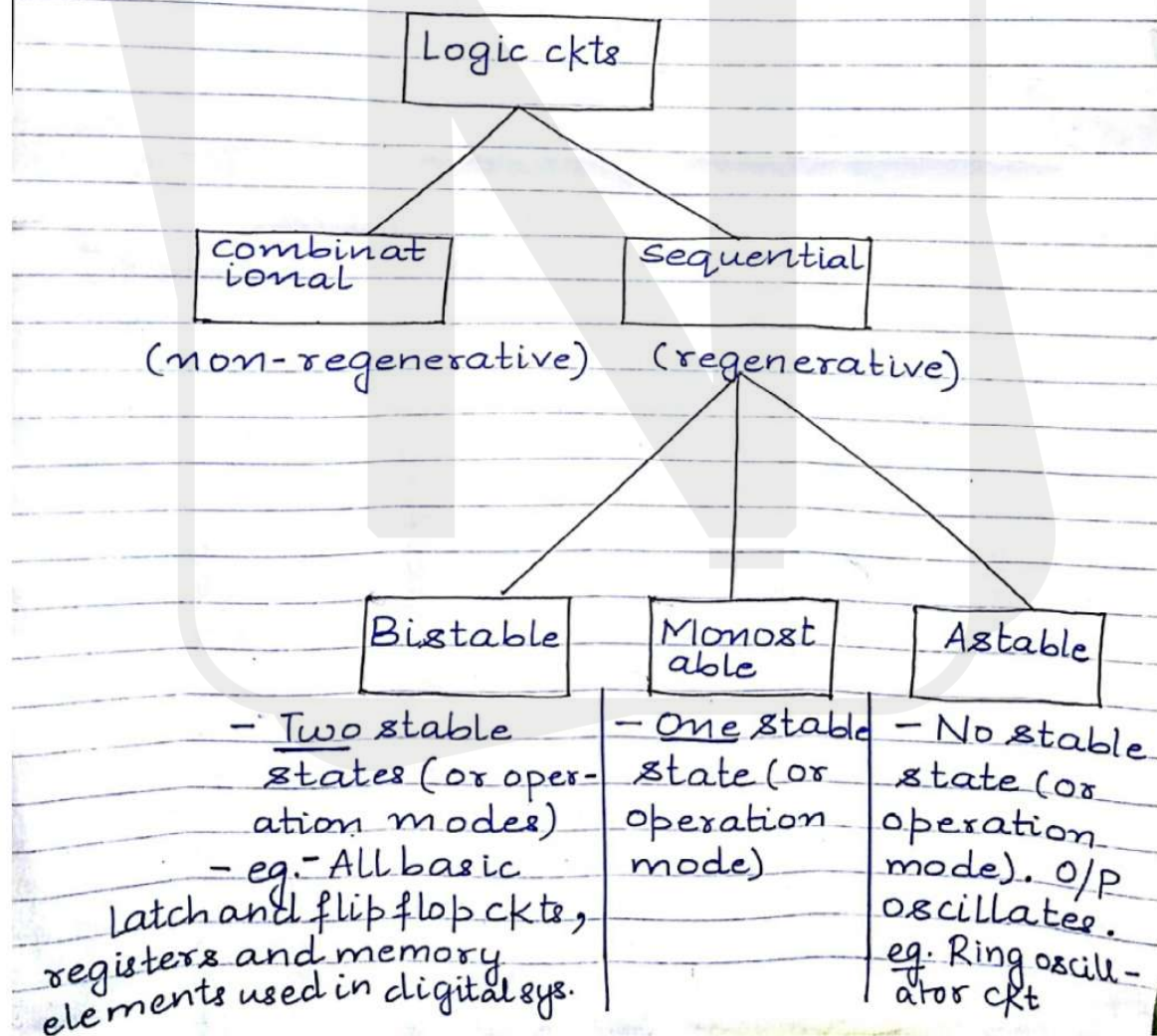
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Classification of logic ckt



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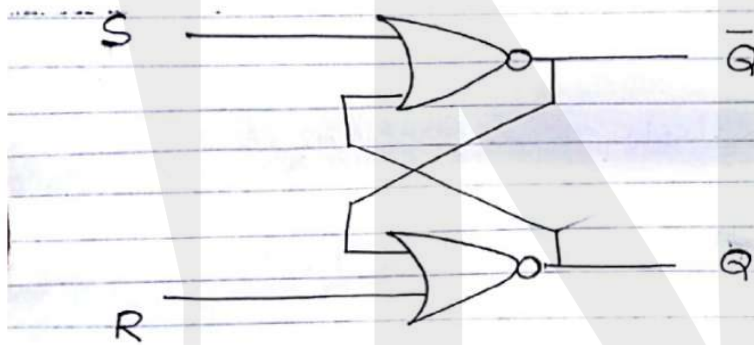
(3)

SR latch ckt

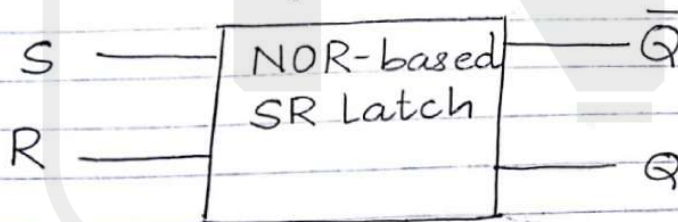
→ A latch is an example of bistable multivibrator that is, a device with exactly two stable states. These states are high-output and low-output. A latch has a feedback path, so information can be retained by the device.

① SR latch using NOR gates

→ Gate-level schematic/ Cross-coupled NOR gates



→ Block diagram



→ A latch is said to be set when Q is equal to logic '1' and $\bar{Q}$ is equal to logic '0'.

→ latch is said to be reset when Q is equal to logic '0' and $\bar{Q}$ is equal to logic '1'.

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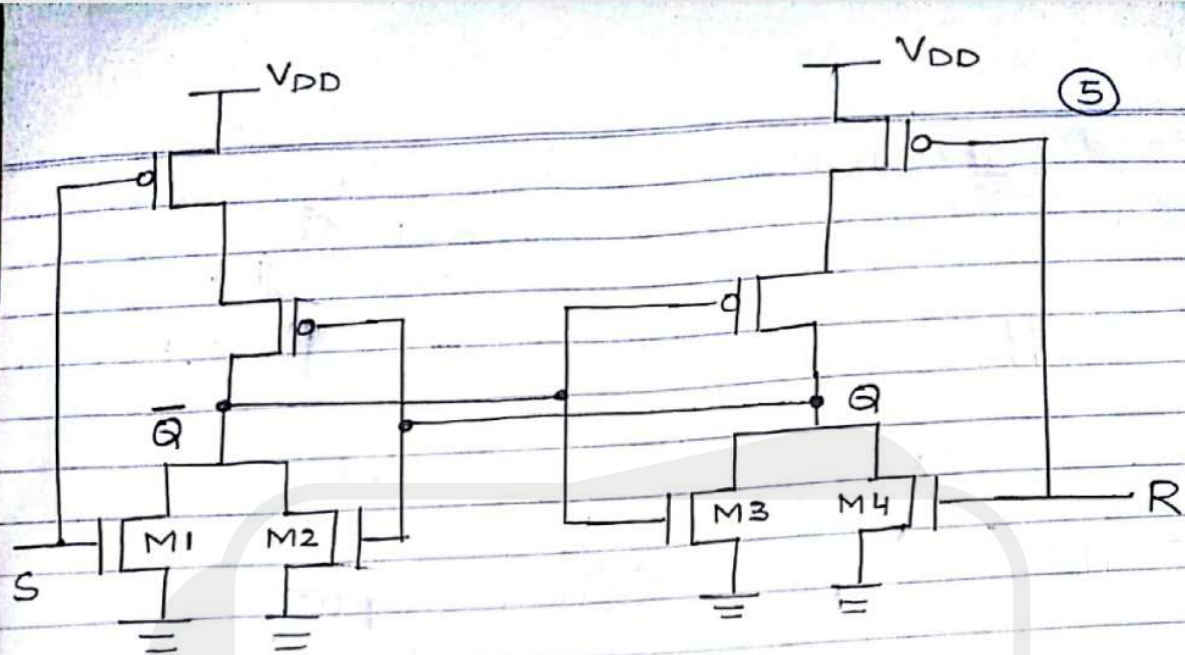
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→ Truth table

S	R	Q_{n+1}	$\overline{Q_{n+1}}$	Operation
0	0	Q_n	$\overline{Q_n}$	Hold
0	1	0	1	Reset
1	0	1	0	Set
1	1	0	0	Not allowed

→ CMOS SR latch (2.5 marks)

nMOS is in parallel
pMOS is in series V_{DD}



→ Operation modes of the transistors in the NOR based CMOS SR latch

S	R	Q_{n+1}	$\overline{Q}_{n+1}$	Operation
V_{OH}	V_{OL}	V_{OH}	V_{OL}	M1 and M2 on, M3 and M4 off
V_{OL}	V_{OH}	V_{OL}	V_{OH}	M1 and M2 off, M3 and M4 on
V_{OL}	V_{OL}	V_{OH}	V_{OL}	M1 and M4 off, M2 on, or
V_{OL}	V_{OL}	V_{OL}	V_{OH}	M1 and M4 off, M3 on

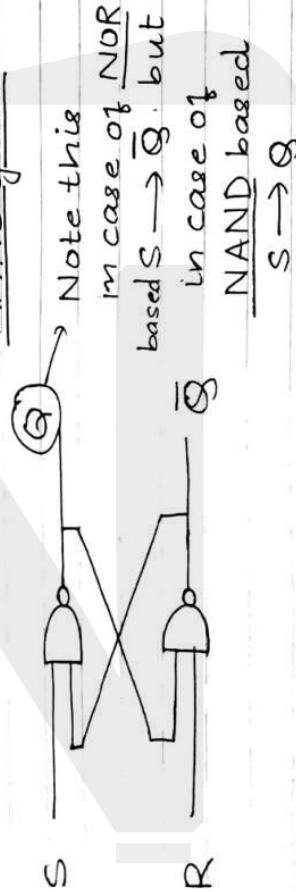
→ If Set and Reset are low, both M1 and M4 are off, and the latch holds its existing state.

→ Depletion-load nMOS SR Latch based on NOR2 gates



● SR Latch using NAND gate

→ Gate-level schematic / cross-coupled NAND gate



→ Block diagram

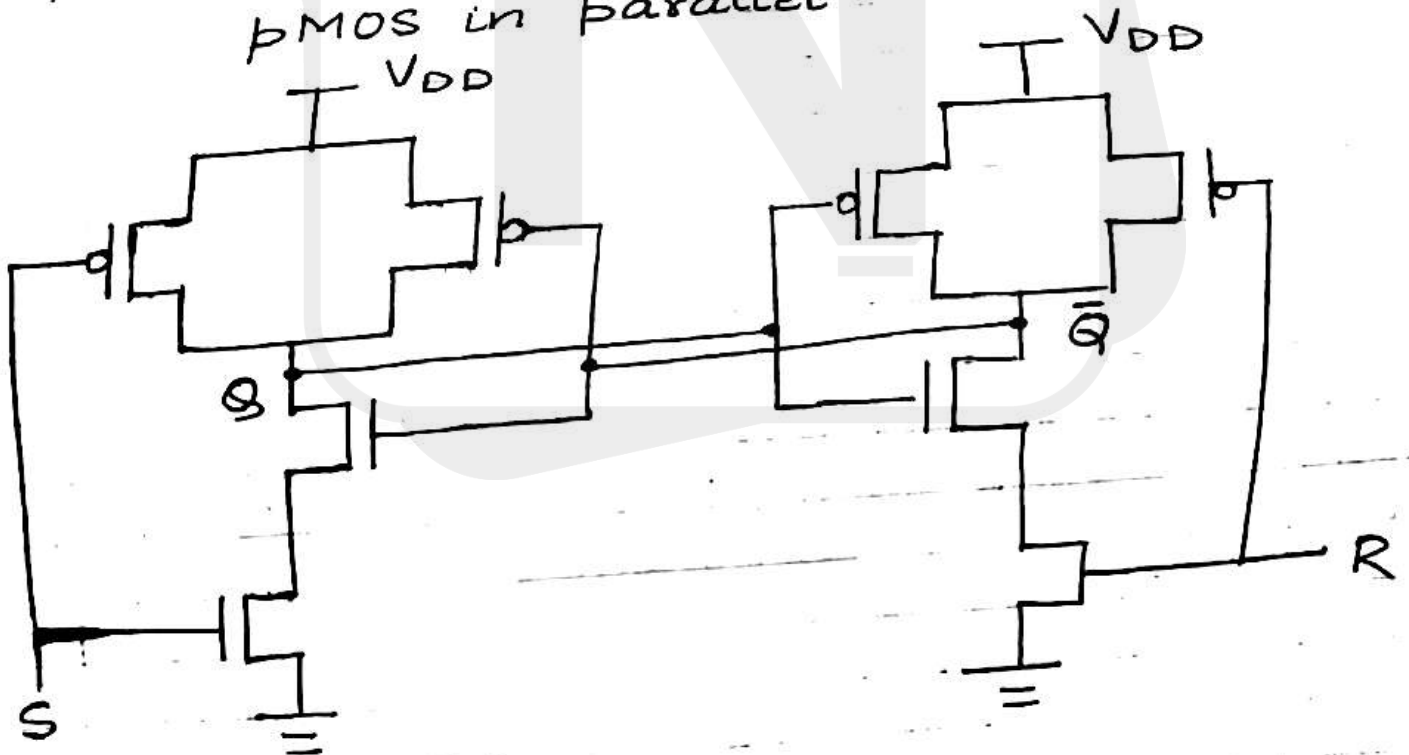


→ Truth table

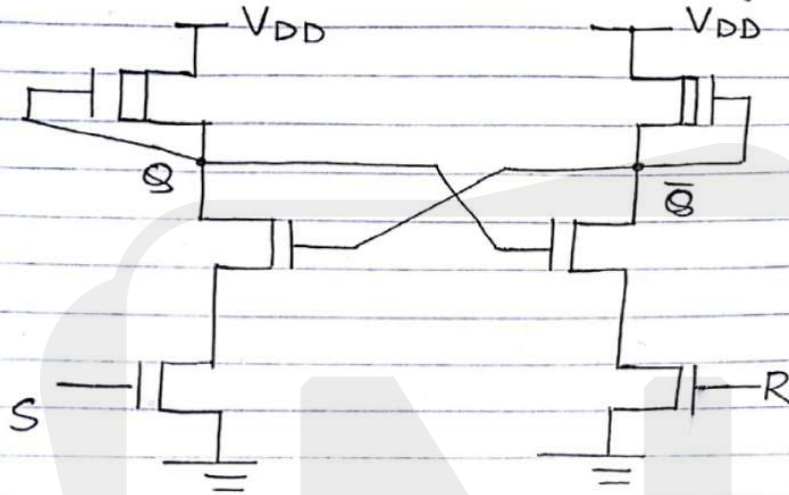
S	R	Q_{n+1}	$\overline{Q}_{n+1}$	Operation
0	0	1	1	not allowed
0	1	1	0	set
1	0	0	1	Reset
1	1	Q_n	$\overline{Q}_n$	hold

→ CMOS SR latch ckt (4 marks)

nMOS in series
pMOS in parallel



⑧
→ Depletion-load nMOS SR latch ckt
based on NOR AND 2 gates



= CMOS ckt implementation is a better alternative ~~be~~ in terms of static power dissipation and noise margin.

= NAND based SR latch responds to active low i/p signals and NOR-based SR latch responds to active high i/p signals.

CLOCKED LATCH & FLIP FLOP _{ckts}

→ Latches are asynchronous sequential circuits, which will respond to changes occurring in input signals at a circuit-delay-dependent time point during their operation.

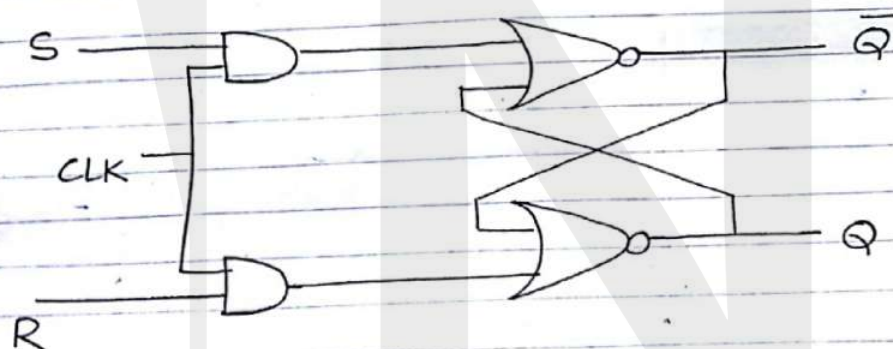
(9)

→ Clocked latch are synchronous sequential ckt's in which the output will respond to the input level only during the active period of a clock pulse.

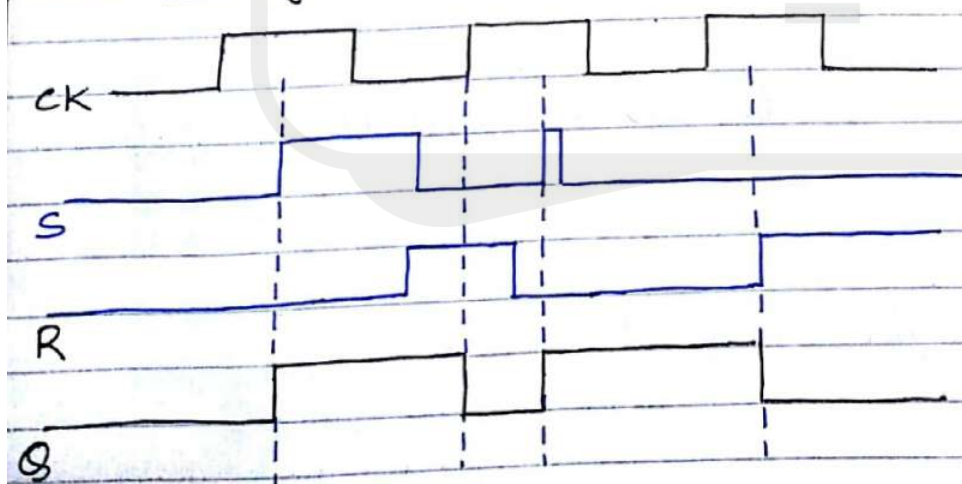
→ Clock pulse is a periodic square waveform.

⊙ Clocked NOR-based SR Latch

→ Gate level schematic



→ Sample i/p and o/p waveforms illustrating the operation (Ckt is strictly level-sensitive)



(AND-OR Invert) $\rightarrow$ sum of product realization (10)

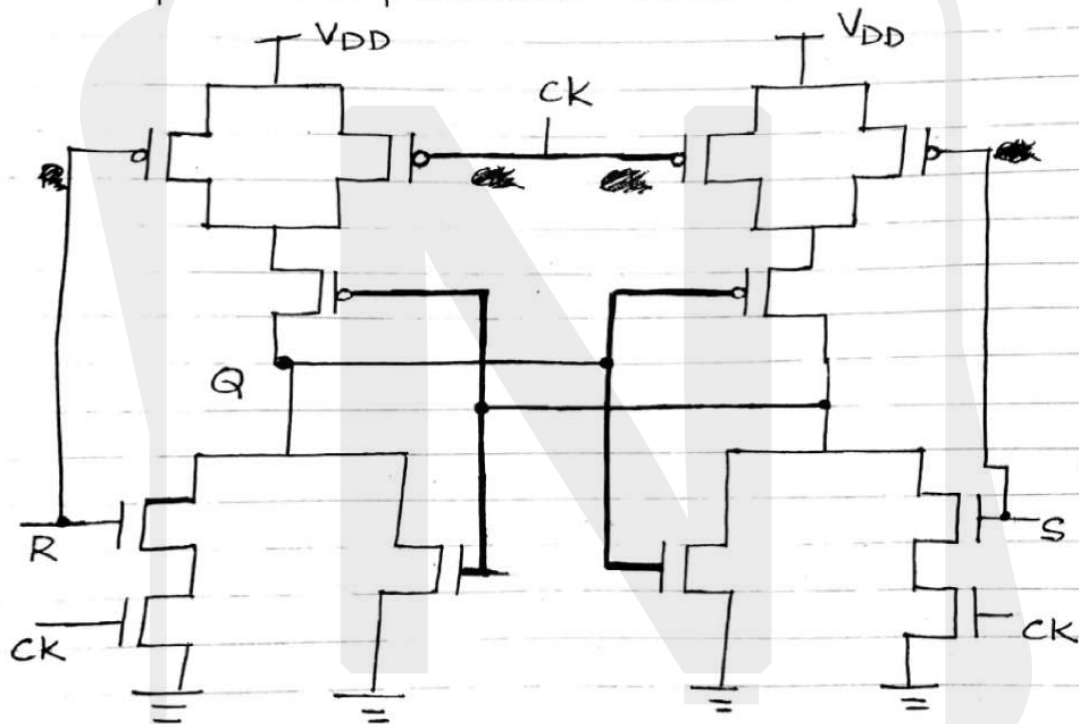
$\rightarrow$ AOI-based implementation of the clocked NOR based SR latch ckt

nMOS is series — AND

nMOS is parallel — OR

pMOS is series — OR

pMOS is parallel — AND



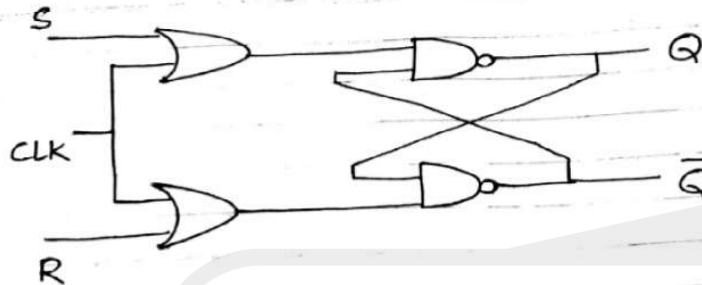
● Clocked SR latch based on NAND gate

(a) ACTIVE LOW INPUTS

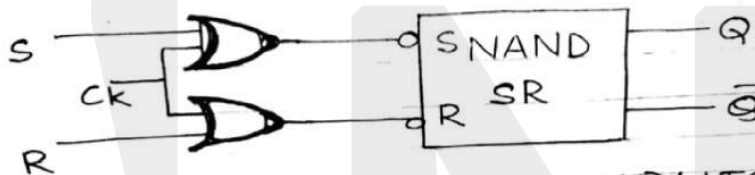
$\left[\begin{array}{l} S = '0' \text{ (activated)} \\ R = '0' \text{ (activated)} \end{array} \right]$
when CK=0 it responds

$\rightarrow$ Gate level schematic

(11)



→ Partial block diag

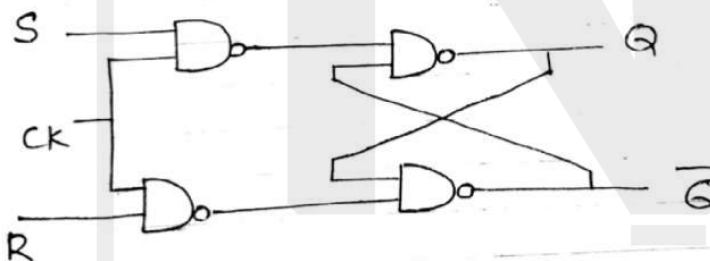


$CK=1$ $S=0$ $R=1$
 $Q=0$

ACTIVE HIGH INPUTS

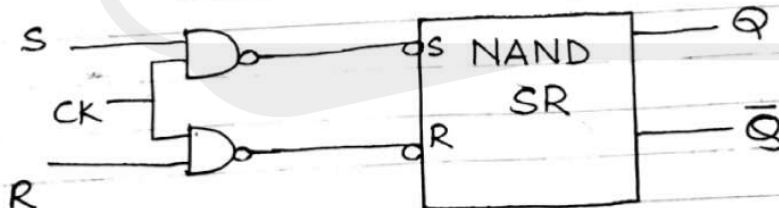
$S=1$ and $R=0$, $CK=1$
 $\Rightarrow Q=1$ (activated)

→ Gate-level schematic



→ Here the transistor count is higher.

→ Partial block diag

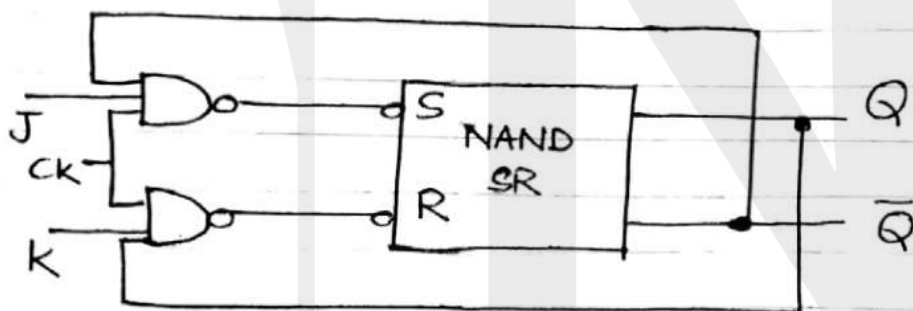


→ The SR Latch has a problem in that when both S and R are high, its states becomes indeterminate

→ The JK Latch eliminates the problem by using feedback from output to input, such that all states in the truth table are allowable.

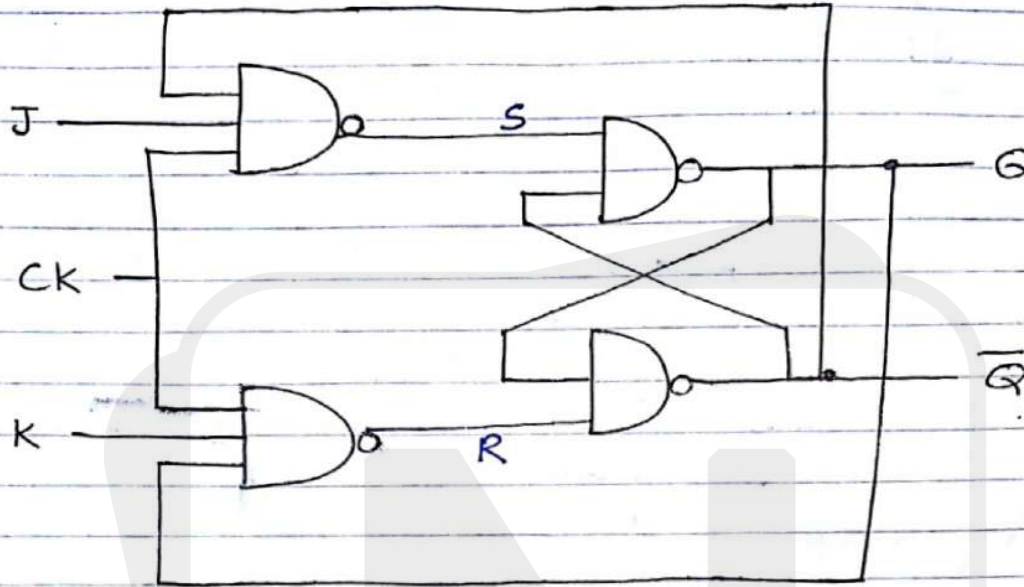
● NAND-based JK Latch

→ Gate level schematic of the clocked NAND-based JK latch ckt



(13)

→ ALL NAND implementation of the clocked JK latch ckt



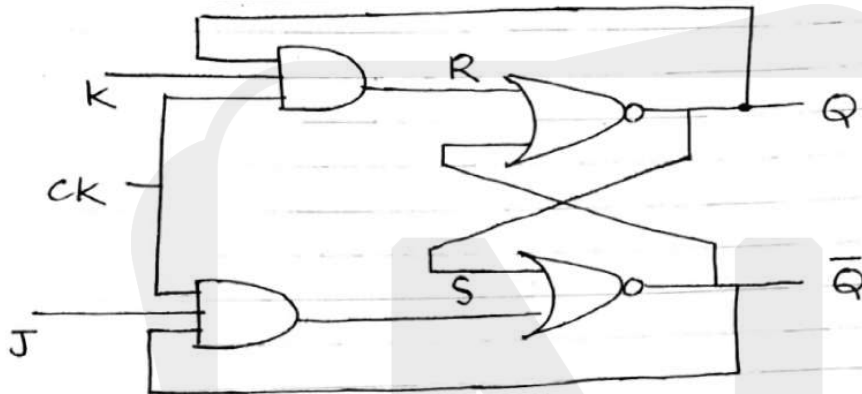
→ Detailed truth table of JK latch ckt
CK=1

J	K	Q_n	Q_n	S	R	Q_{n+1}	Q_{n+1}	Operat ⁿ
0	0	0	1	1	1	0	1	hold
		1	0	1	1	1	0	
0	1	0	1	1	1	0	1	reset
		1	0	1	0	0	1	
1	0	0	1	0	1	1	0	set
		1	0	1	1	1	0	
1	1	0	1	0	1	1	0	toggle
		1	0	1	0	0	1	

(14)

① NOR-based JK latch

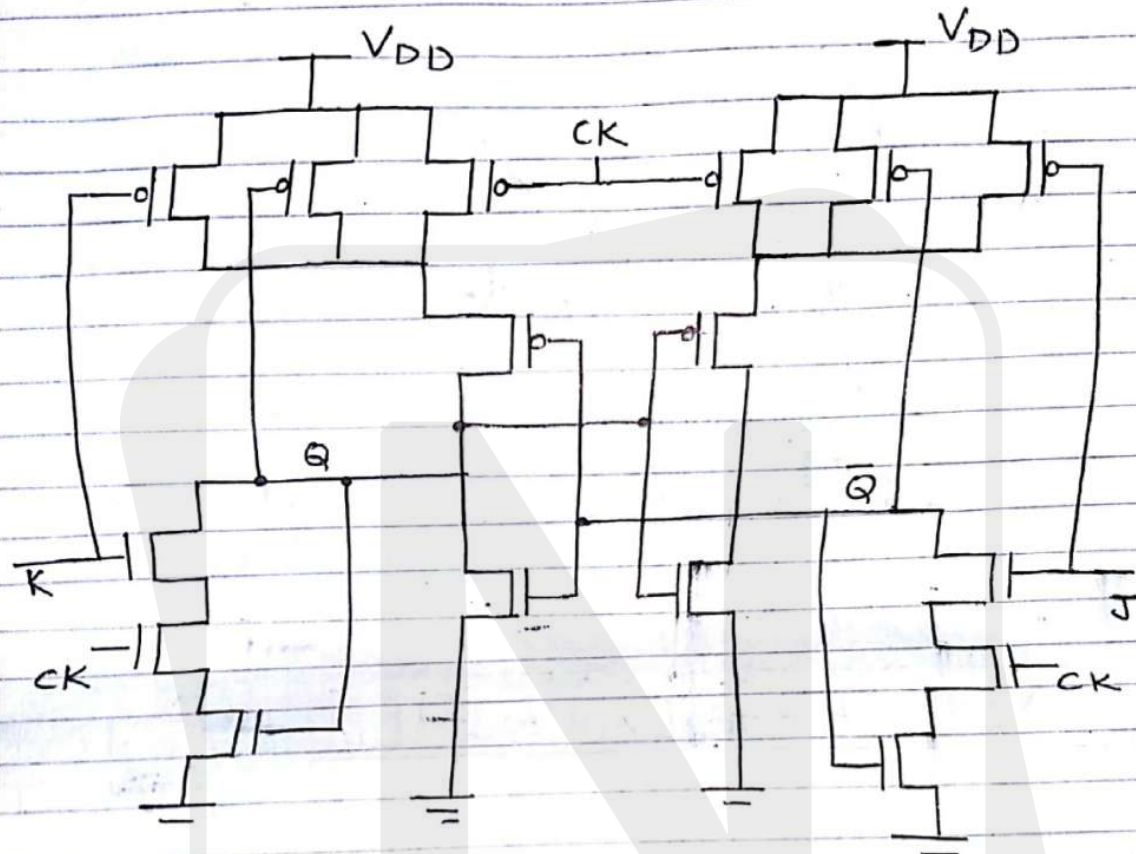
→ Gate level schematic of the clocked NOR-based JK Latch



→ CMOS AOI realization of the JK latch
(7 marks)

nmos is in series - AND
 " " " parallel - OR
 pmos is in series - OR
 " " " parallel - AND

(15)



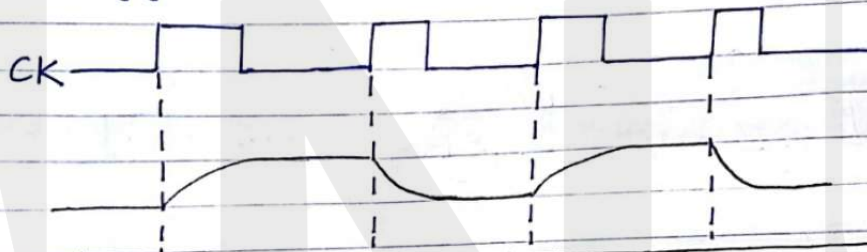
→ Problem with JK latch

When both inputs are equal to logic 1^{“6”} during the active phase of the clock pulse, the o/p of the circuit will oscillate (toggle) until the clock becomes inactive (goes to zero), one of the i/p signals goes to zero. To solve this clock pulse width should be made smaller than i/p-o/p propagation delay.

(16)

However this clock constraint is difficult to implement for practical application.

→ If the o/p of the JK latch toggle (change its state) only once for each clock pulse, when both inputs are equal to logic 1, A circuit which is operated in this mode is called a toggle switch.



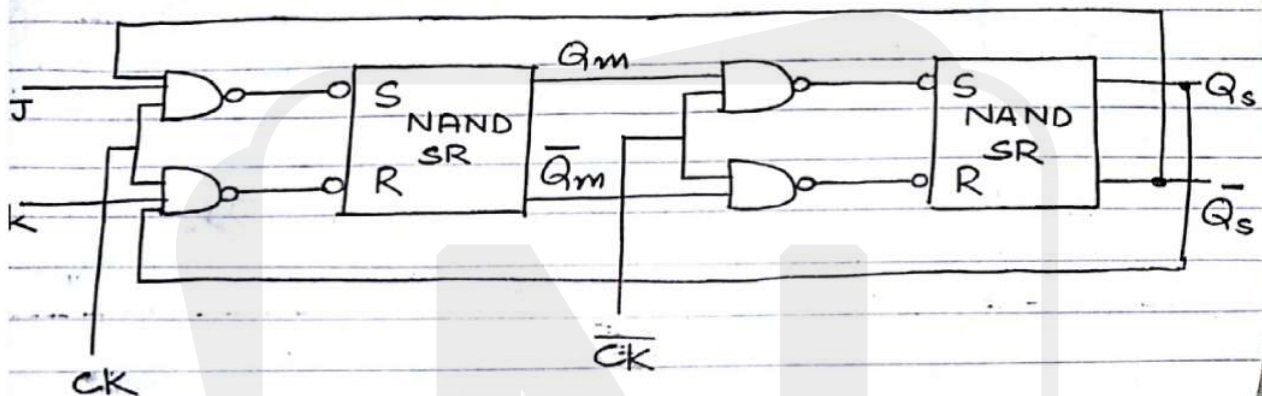
MASTER-SLAVE FLIP FLOP

is defined as two latches connected serially and activated with opp. phase clocks

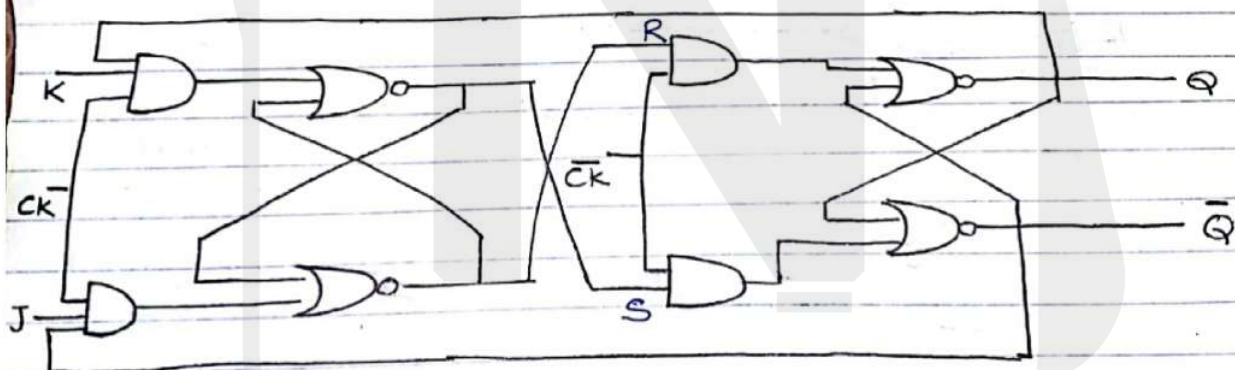
- First latch is the Master; Second latch is the slave
- Eliminates transparency, i.e. a change occurring in the primary input is never reflected directly to the o/p, since opposite phase clocks are used to activate M and S latches.

(17)

→ Gate level schematic of master-slave flip flop consisting of NAND-based JK latches.



→ NOR based realization of JK master-slave flip-flop



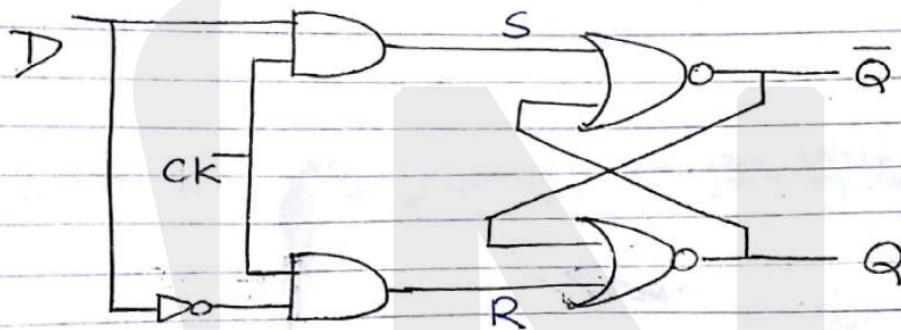
↳ Problem

When the clock pulse is high, a narrow spike or glitch in one of the i/p's, for instance a glitch in the J line (or K line), may set (or reset) the master latch and thus cause an unwanted state transition, which will be propagated into the slave stage. This is called

ones catching'. To solve this edge trigger master-slave flip flop should be used.

CMOS D-LATCH & EDGE-TRIGGER - RED FLIP FLOP

↳ Gate level schematic

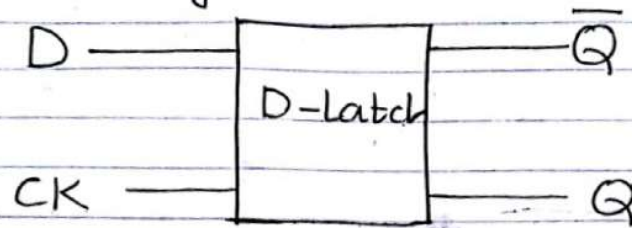


— A D-latch is implemented, at the gate level, by simply utilizing a NOR-based S-R Latch connecting D to input S, and connecting D' to input R with an inverter.

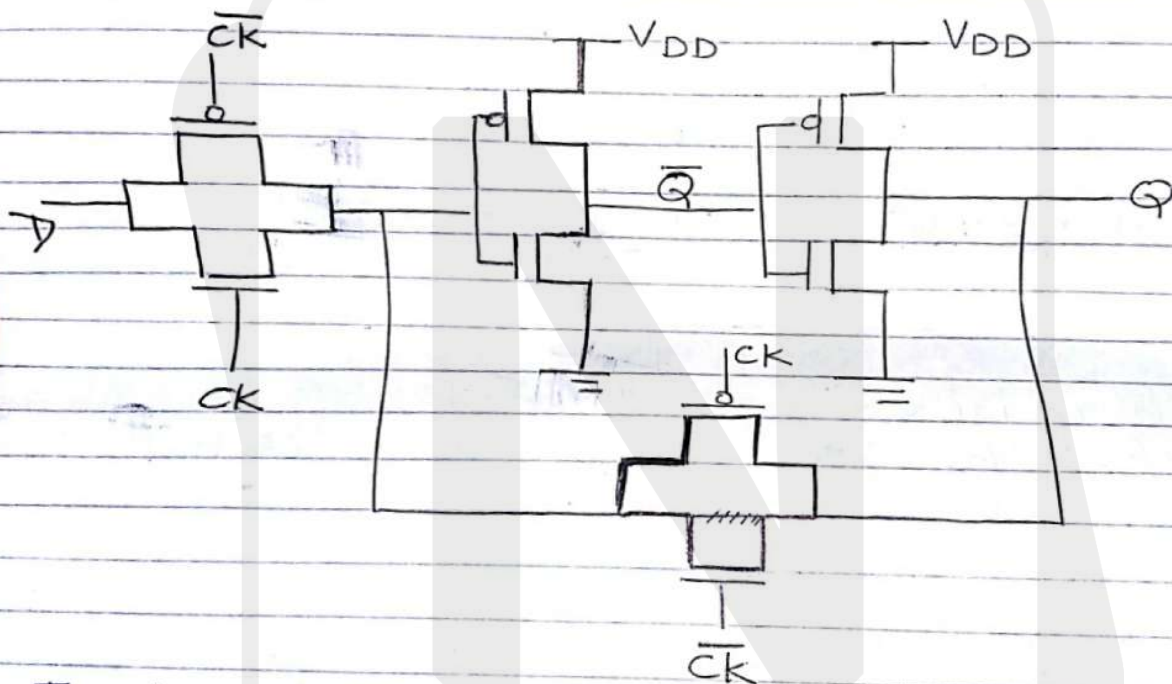
— When clk goes high, D is transmitted to output Q (and D' to Q').
When clk goes low, the latch retains its previous state.

(19)

→ Block diag



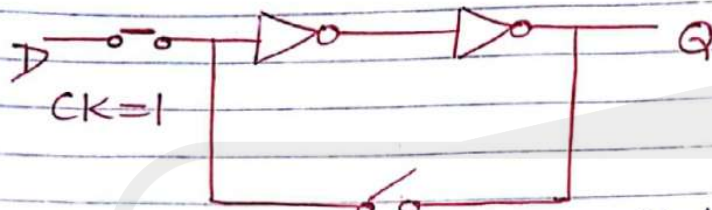
→ CMOS implementation of the D-latch (version 1)



The input TG is activated with CLK while the latch feedback loop TG is activated with CLK'. Input D is accepted when CLK is high. When CLK goes low, the input is open-ckted and the latch is set with the prior data D.

(20)

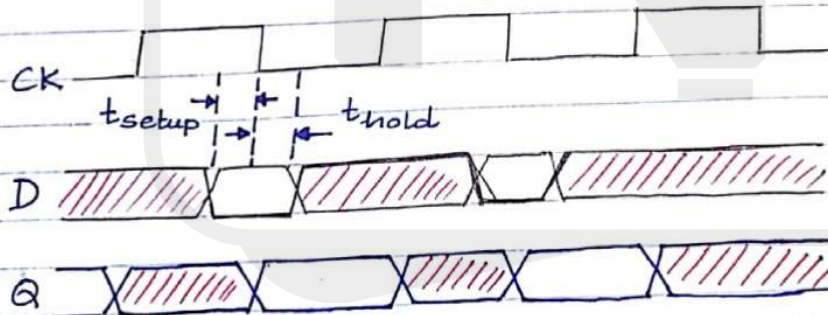
→ Schematic view and timing diag



When $CK=1$, the i/p switch is closed allowing new data into the latch.



When $CK=0$, the i/p switch is opened and the feedback loop switch is closed, setting the latch.



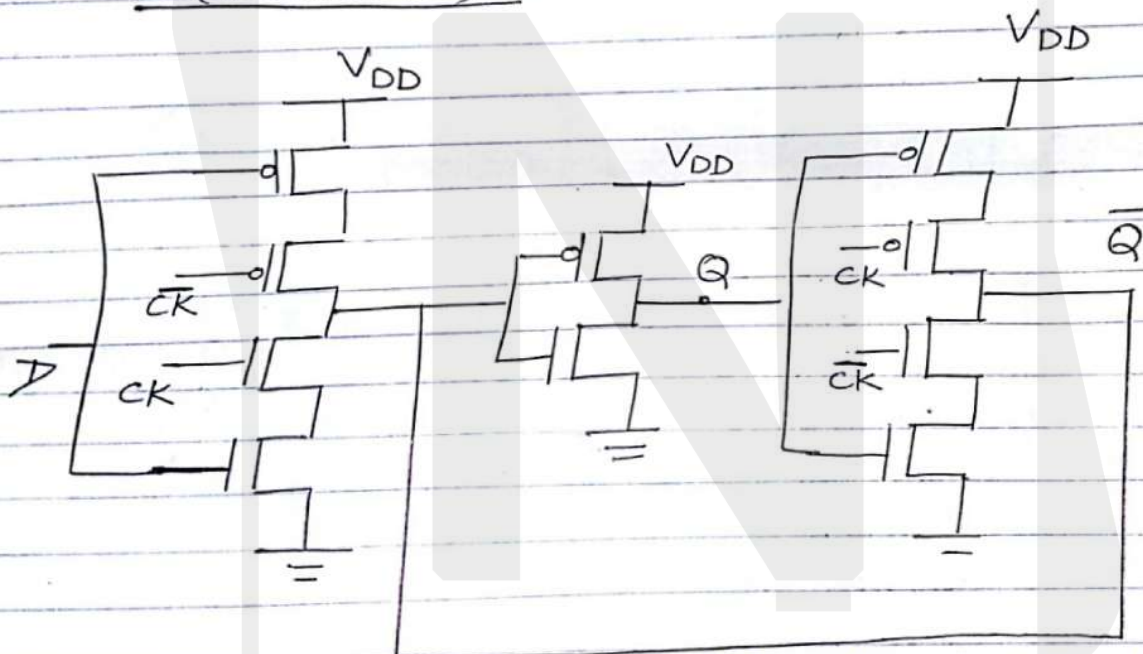
t_{setup} : - Time upto which the i/p (D) must remain stable before clock transition

(21)

t_{hold} : time upto which the i/p must remain stable after clock transition

Violation of t_{setup} and t_{hold} can cause metastability problems and chaotic transient behaviour.

→ CMOS implementation of D latch
(version 2)



When CLK is high, the first tri-state inverter sends the inverted i/p through the second inverter, while the second tri-state is in its high Z state.

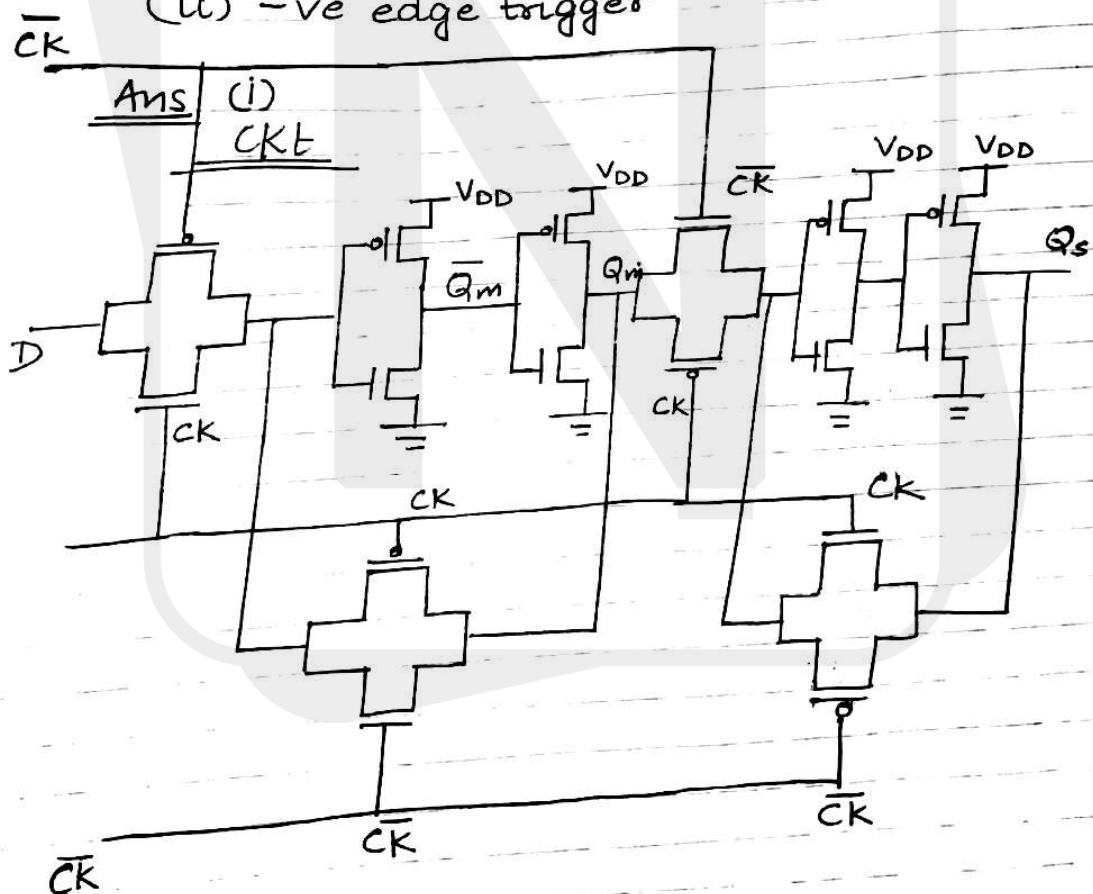
O/p Q is following i/p D.

(22)

When CLK is low, the first tri-state goes into its high Z state, while the second tri-state inverter closes the feedback loop, holding the Q and Q' in the latch.

~~The~~ The ckt's discussed till now are level triggered.

Q Implement CMOS, D latch for the following cases —
 (i) +ve edge trigger
 (ii) -ve edge trigger



Explanation

- The master is negative-level sensitive while the slave is positive-level sensitive
- When the clock is low, $\uparrow$ the master follows the D input while the slave holds the previous value. ($Q_m = D'$)
- When the clock changes from logic '0' to logic '1', the master latch ceases to sample the input and stores the D value at the time of the clock transition. At the same time, the slave latch becomes transparent, passing the value stored in the master ($Q_m = D'$) to the o/p of the slave stage, Q_s . The input cannot effect the output because the master stage is disconnected from the D i/p.
- When the clock changes again from 1 to 0, the slave forms a feedback loop, saving the value of master and the open master start to sampling the i/p D again. and following
- This circuit is a positive edge trigger D flip flop by virtue of the fact that it samples the i/p at the rising edge of clock pulse.

